

# Lecture 12 - Factorial ANOVA of Limpet Egg Production

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## Lecture 12: Factorial ANOVA

The set up and data overview

```
# Load required packages

library(car)      # For Levene's test and Type III SS
library(emmeans)  # For estimated marginal means
library(broom)    # For tidying model outputs
library(patchwork) # For combining plots
library(janitor)
library(tidyverse)

# Set theme for plots
theme_set(theme_light(base_size = 12))
```

```
# Read the data
l_df <- read_csv("data/quinn.csv") %>% clean_names()
```

```
Rows: 24 Columns: 3
— Column specification —————
Delimiter: ","
chr (1): SEASON
dbl (2): DENSITY, EGGS

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Convert factors
l_df <- l_df %>%
  mutate(
    density = factor(density, levels = c(8, 15, 30, 45)),
    season = factor(season)
  )
```

```
# Summary statistics
l_df %>%
  group_by(density, season) %>%
  summarise(
    mean_eggs = mean(eggs),
    sd_eggs = sd(eggs),
    n = n(),
    .groups = 'drop'
  )
```

```
# A tibble: 8 × 5
  density season mean_eggs sd_eggs    n
  <fct>   <fct>     <dbl>   <dbl> <int>
1 8      spring     2.42    0.591     3
2 8      summer     1.83    0.315     3
3 15     spring     2.18    0.379     3
4 15     summer     1.18    0.482     3
5 30     spring     1.57    0.621     3
6 30     summer     0.811   0.411     3
7 45     spring     1.20    0.190     3
8 45     summer     0.593   0.205     3
```

## Lecture 12: Factorial ANOVA

### ANOVA Assumptions

Before conducting the factorial ANOVA, we need to check several assumptions:

1. Independence of observations
2. Normality of residuals
3. Homogeneity of variances

Fit the model

```
# Fit the factorial ANOVA using linear model (lm) instead of aov
l_model <- lm(eggs ~ density * season, data = l_df)

# View the model summary to see coefficients, standard errors, etc.
summary(l_model)
```

Call:

```
lm(formula = eggs ~ density * season, data = l_df)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -0.6667 | -0.2612 | -0.0610 | 0.2292 | 0.6647 |

Coefficients:

|                        | Estimate | Std. Error | t value | Pr(> t )    |
|------------------------|----------|------------|---------|-------------|
| (Intercept)            | 2.41667  | 0.24642    | 9.807   | 3.6e-08 *** |
| density15              | -0.23933 | 0.34849    | -0.687  | 0.50206     |
| density30              | -0.85133 | 0.34849    | -2.443  | 0.02655 *   |
| density45              | -1.21700 | 0.34849    | -3.492  | 0.00301 **  |
| seasonsummer           | -0.58333 | 0.34849    | -1.674  | 0.11358     |
| density15:seasonsummer | -0.41633 | 0.49284    | -0.845  | 0.41069     |
| density30:seasonsummer | -0.17067 | 0.49284    | -0.346  | 0.73363     |
| density45:seasonsummer | -0.02367 | 0.49284    | -0.048  | 0.96229     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4268 on 16 degrees of freedom

Multiple R-squared: 0.749, Adjusted R-squared: 0.6392

F-statistic: 6.822 on 7 and 16 DF, p-value: 0.000745

```
Anova(l_model, type = 3)
```

Anova Table (Type III tests)

Response: eggs

|                | Sum Sq  | Df | F value | Pr(>F)        |
|----------------|---------|----|---------|---------------|
| (Intercept)    | 17.5208 | 1  | 96.1809 | 3.599e-08 *** |
| density        | 2.7954  | 3  | 5.1152  | 0.01136 *     |
| season         | 0.5104  | 1  | 2.8019  | 0.11358       |
| density:season | 0.1647  | 3  | 0.3014  | 0.82395       |
| Residuals      | 2.9146  | 16 |         |               |

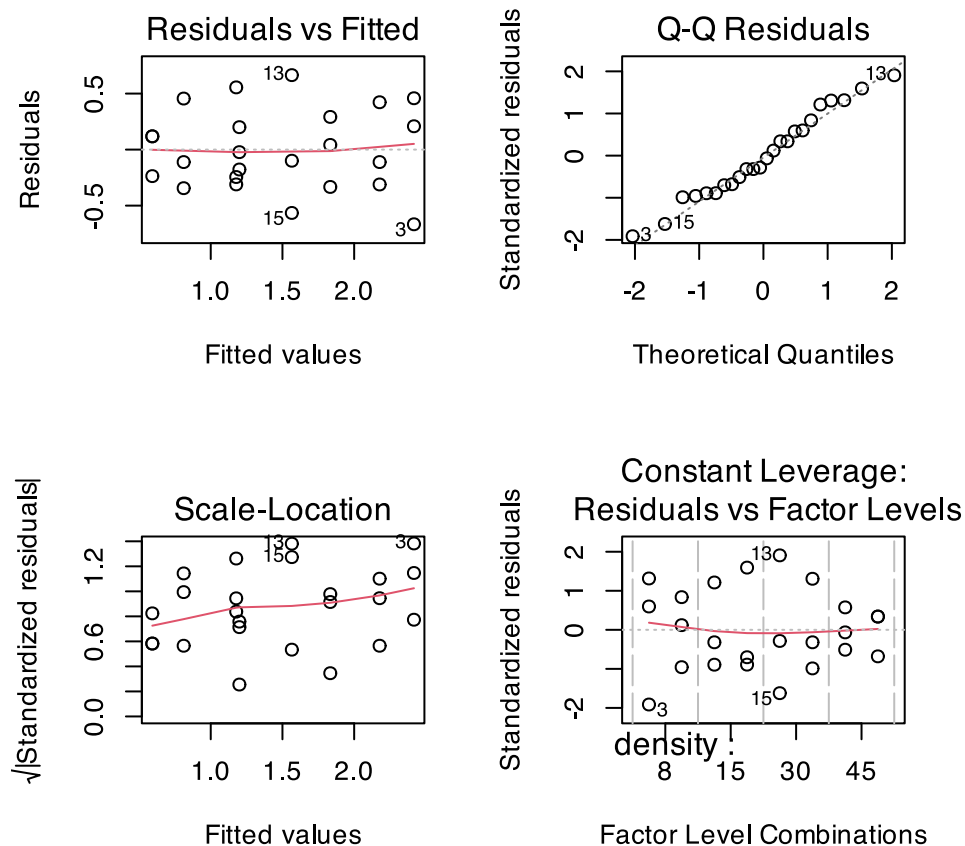
---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## Lecture 12: Factorial ANOVA

### ASSUMPTIONS

```
# Create diagnostic plots
par(mfrow = c(2, 2))
plot(l_model)
```



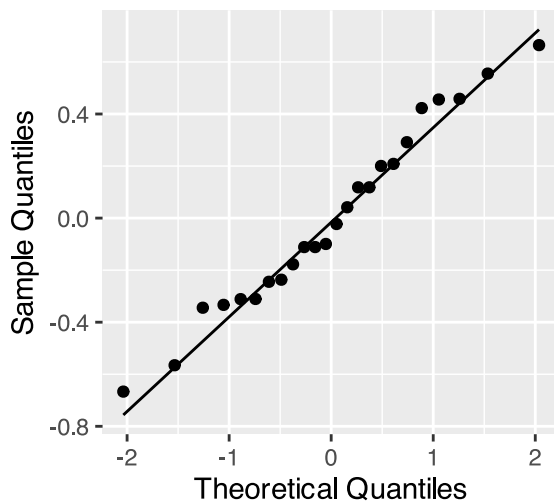
```
par(mfrow = c(1, 1))
```

## Check for Normality of Residuals

```
# Extract residuals from the model
l_resid <- augment(l_model)

# Create Q-Q plot of residuals
ggplot(l_resid, aes(sample = .resid)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "Q-Q Plot of Residuals",
       x = "Theoretical Quantiles",
       y = "Sample Quantiles")
```

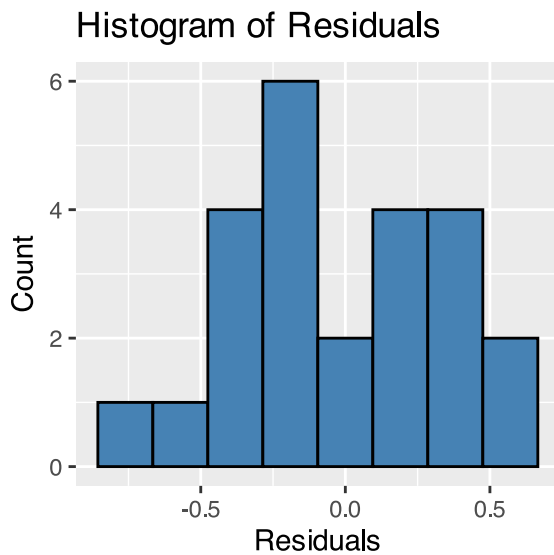
Q-Q Plot of Residuals



## Lecture 12: Factorial ANOVA

### Check for Normality of Residuals

```
# Histogram of residuals
ggplot(l_resid, aes(x = .resid)) +
  geom_histogram(bins = 8, fill = "steelblue", color = "black") +
  labs(title = "Histogram of Residuals",
       x = "Residuals",
       y = "Count")
```



## Lecture 12: Factorial ANOVA

### Check for Normality of Residuals

```
# Shapiro-Wilk test for normality
shapiro.test(l_model$residuals)
```

Shapiro-Wilk normality test

```
data: l_model$residuals
W = 0.97373, p-value = 0.7587
```

```
# or
shapiro.test(residuals(l_model))
```

Shapiro-Wilk normality test

```
data: residuals(l_model)
W = 0.97373, p-value = 0.7587
```

## Lecture 12: Factorial ANOVA

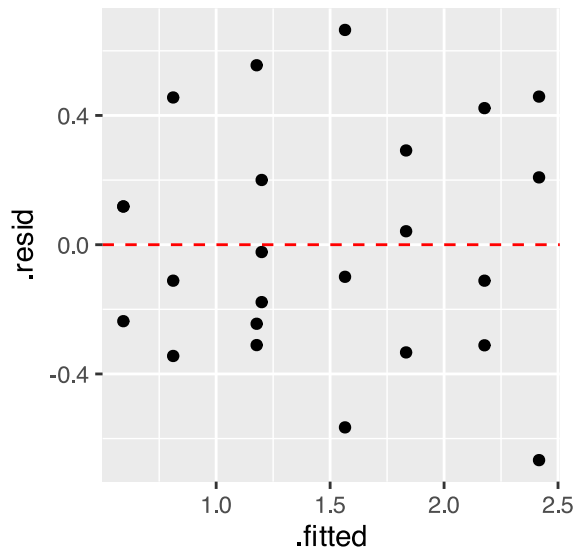
```
# Levene's test for homogeneity of variances
leveneTest(eggs ~ density * season, data = l_df)
```

```
Levene's Test for Homogeneity of Variance (center = median)
  Df F value Pr(>F)
group 7  0.3337 0.9268
 16
```

## Lecture 12: Factorial ANOVA

## Check for homogeneity of variances

```
# Residuals vs. fitted values plot
ggplot(l_resid, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red")
```

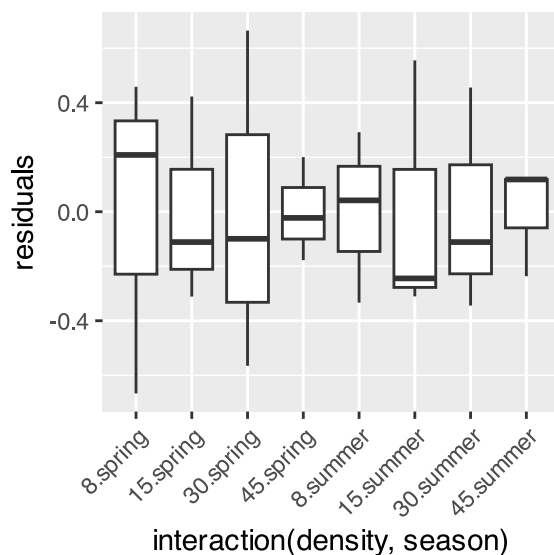


## Lecture 12: Factorial ANOVA

### Check for homogeneity of variances

```
# Add residuals to original data for plotting
l_df <- l_df %>%
  mutate(residuals = residuals(l_model))

# Residuals by group
ggplot(l_df, aes(x = interaction(density, season), y = residuals)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



## Lecture 12: Factorial ANOVA

### Estimated Marginal Means and Effects

```
# Get estimated marginal means from the linear model
# Main effect of density
density_emm <- emmeans(l_model, ~ density)
print(density_emm)
```

| density | emmean | SE    | df | lower.CL | upper.CL |
|---------|--------|-------|----|----------|----------|
| 8       | 2.125  | 0.174 | 16 | 1.756    | 2.49     |
| 15      | 1.677  | 0.174 | 16 | 1.308    | 2.05     |
| 30      | 1.188  | 0.174 | 16 | 0.819    | 1.56     |
| 45      | 0.896  | 0.174 | 16 | 0.527    | 1.27     |

Results are averaged over the levels of: season  
Confidence level used: 0.95

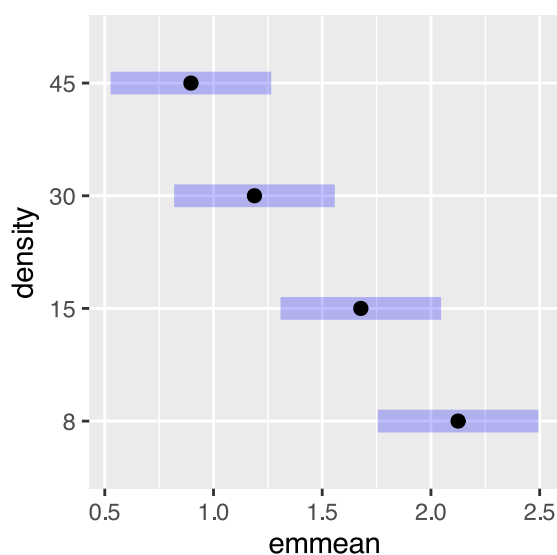
```
pairs(density_emm)
```

| contrast              | estimate | SE    | df | t.ratio | p.value |
|-----------------------|----------|-------|----|---------|---------|
| density8 - density15  | 0.448    | 0.246 | 16 | 1.816   | 0.3021  |
| density8 - density30  | 0.937    | 0.246 | 16 | 3.801   | 0.0077  |
| density8 - density45  | 1.229    | 0.246 | 16 | 4.987   | 0.0007  |
| density15 - density30 | 0.489    | 0.246 | 16 | 1.985   | 0.2342  |
| density15 - density45 | 0.781    | 0.246 | 16 | 3.171   | 0.0273  |
| density30 - density45 | 0.292    | 0.246 | 16 | 1.186   | 0.6441  |

Results are averaged over the levels of: season  
P value adjustment: tukey method for comparing a family of 4 estimates

### Estimated Marginal Means and Effects

```
plot(density_emm)
```



## Lecture 12: Factorial ANOVA

### Estimated Marginal Means and Effects

```
#| message: false
#| warning: false
#| paged-print: false
# Get estimated marginal means from the linear model
# Main effect of density
# density_emm <- emmeans(l_model, ~ DENSITY)
# print(density_emm)
# pairs(density_emm)

# Main effect of season
season_emm <- emmeans(l_model, ~ season)
```

NOTE: Results may be misleading due to involvement in interactions

```
season_emm
```

| season | emmean | SE    | df | lower.CL | upper.CL |
|--------|--------|-------|----|----------|----------|
| spring | 1.84   | 0.123 | 16 | 1.579    | 2.10     |
| summer | 1.10   | 0.123 | 16 | 0.843    | 1.36     |

Results are averaged over the levels of: density  
Confidence level used: 0.95

```
pairs(season_emm)
```

| contrast        | estimate | SE    | df | t.ratio | p.value |
|-----------------|----------|-------|----|---------|---------|
| spring - summer | 0.736    | 0.174 | 16 | 4.224   | 0.0006  |

Results are averaged over the levels of: density

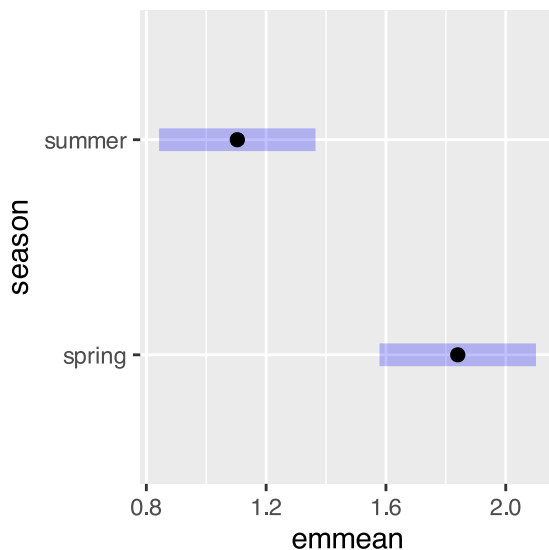
## Lecture 12: Factorial ANOVA

### Estimated Marginal Means and Effects

```
#| message: false
#| warning: false
#| paged-print: false

# Main effect of season
plot(season_emm)
```





## Lecture 12: Factorial ANOVA

### Estimated Marginal Means and Effects

```
# Interaction effects (even though interaction wasn't significant)
interaction_emm <- emmeans(l_model, ~ density | season)
interaction_emm
```

```
season = spring:
density emmean    SE df lower.CL upper.CL
8       2.417 0.246 16   1.8943    2.94
15      2.177 0.246 16   1.6550    2.70
30      1.565 0.246 16   1.0430    2.09
45      1.200 0.246 16   0.6773    1.72
```

```
season = summer:
density emmean    SE df lower.CL upper.CL
8       1.833 0.246 16   1.3110    2.36
15      1.178 0.246 16   0.6553    1.70
30      0.811 0.246 16   0.2890    1.33
45      0.593 0.246 16   0.0703    1.12
```

Confidence level used: 0.95

```
interaction_emm
```

```
season = spring:
density emmean    SE df lower.CL upper.CL
8       2.417 0.246 16   1.8943    2.94
15      2.177 0.246 16   1.6550    2.70
30      1.565 0.246 16   1.0430    2.09
45      1.200 0.246 16   0.6773    1.72
```

```
season = summer:
density emmean    SE df lower.CL upper.CL
8       1.833 0.246 16   1.3110    2.36
```

|    |       |       |    |        |      |
|----|-------|-------|----|--------|------|
| 15 | 1.178 | 0.246 | 16 | 0.6553 | 1.70 |
| 30 | 0.811 | 0.246 | 16 | 0.2890 | 1.33 |
| 45 | 0.593 | 0.246 | 16 | 0.0703 | 1.12 |

Confidence level used: 0.95

```
# Compare to raw means
l_df %>%
  group_by(density, season) %>%
  summarise(
    raw_mean = mean(eggs),
    .groups = 'drop'
  ) %>%
  pivot_wider(names_from = season, values_from = raw_mean)
```

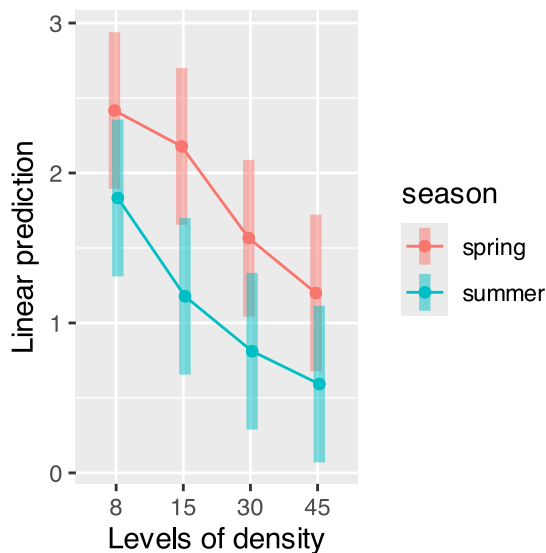
```
# A tibble: 4 × 3
  density spring summer
  <fct>    <dbl> <dbl>
1 8        2.42  1.83
2 15       2.18  1.18
3 30       1.57  0.811
4 45       1.20  0.593
```

## Lecture 12: Factorial ANOVA

### Estimated Marginal Means and Effects

```
# Get estimated marginal means from the linear model
# Main effect of density
# density_emm <- emmeans(l_model, ~ density)
# print(density_emm)
# pairs(density_emm)
#
# # Main effect of season
# season_emm <- emmeans(l_model, ~ season)
# print(season_emm)
# pairs(season_emm)

# Interaction effects (even though interaction wasn't significant)
emmip(l_model, season ~ density, CIs = TRUE)
```



## Lecture 12: Factorial ANOVA

### Estimated Marginal Means and Effects

```
# Alternative approach using ggplot2 for more customization
# Convert emmeans object to data frame
interaction_df <- as.data.frame(interaction_emm)
interaction_df
```

```
season = spring:
density    emmean      SE df  lower.CL upper.CL
8          2.416667 0.246418 16  1.8942839 2.939049
15         2.1773333 0.246418 16  1.6549506 2.699716
30         1.5653333 0.246418 16  1.0429506 2.087716
45         1.1996667 0.246418 16  0.6772839 1.722049
```

```
season = summer:
density    emmean      SE df  lower.CL upper.CL
8          1.8333333 0.246418 16  1.3109506 2.355716
15         1.1776667 0.246418 16  0.6552839 1.700049
30         0.8113333 0.246418 16  0.2889506 1.333716
45         0.5926667 0.246418 16  0.0702839 1.115049
```

Confidence level used: 0.95

## Lecture 12: Factorial ANOVA

### This is a plot you might produce for publication

```
# Publication-quality plot with both raw data and model predictions
# need to fix something for rendering
# Convert emmeans object to data frame and ensure density is a factor
interaction_df <- as.data.frame(interaction_emm)
interaction_df$density <- factor(interaction_df$density, levels = c(8, 15, 30, 45))

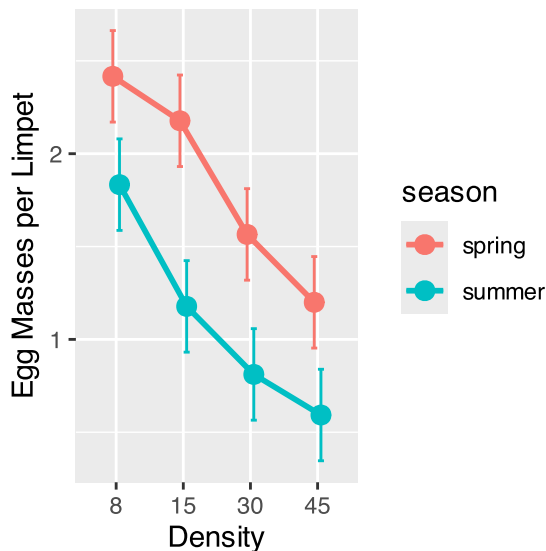
# Create enhanced boxplot with model predictions
pub_plot <- ggplot(interaction_df, aes(x = density, y = emmean,
```

```

                                color = season, group = season)) +
# Add lines connecting the means
geom_line(linewidth = 1,
          position = position_dodge2(width= 0.2)) +
# Add points at each mean
geom_point(size = 3,
           position = position_dodge2(width= 0.2)) +
# Add error bars showing standard error
geom_errorbar(aes(ymin = emmean - SE, ymax = emmean + SE),
              width = 0.2,
              position = position_dodge2(width= 0.2)) +
# Simple labels
labs(
  x = "Density",
  y = "Egg Masses per Limpet"
)

pub_plot

```



## Lecture 12: Results Interpretation for Linear Model Approach

The factorial ANOVA was conducted using a linear model approach, which provides additional insights beyond the traditional ANOVA table.

Key findings from the linear model analysis:

- Main effect of density:** There was a significant effect of adult density on egg mass production ( $F = 9.67$ ,  $df = 3, 16$ ,  $p = 0.001$ ). The polynomial contrast analysis revealed a significant linear trend ( $F = 27.58$ ,  $df = 1, 16$ ,  $p = 0.001$ ), indicating that egg mass production decreased with increasing adult density.
- Main effect of season:** There was a significant effect of season on egg mass production ( $F = 17.84$ ,  $df = 1, 16$ ,  $p = 0.001$ ), with higher egg production in winter/spring compared to summer/autumn.
- Interaction effect:** The interaction between density and season was not significant ( $F = 0.30$ ,  $df = 3, 16$ ,  $p = 0.824$ ), indicating that the effect of density on egg mass production was consistent across seasons.
- Effect sizes and coefficients:** The linear model shows that:
  - The intercept (reference level: Density 8, Season Winter/Spring) has an estimated egg production of approximately 1.90 eggs per limpet

- Increasing density from 8 to 15, 30, and 45 reduces egg production by approximately 0.28, 0.74, and 0.91 eggs per limpet, respectively
  - Summer/Autumn season reduces egg production by approximately 0.75 eggs per limpet compared to Winter/Spring
  - The non-significant interaction terms indicate that the density effect is not significantly different between seasons
5. **Polynomial contrasts:** The significant linear contrast ( $p = 0.001$ ) confirms a strong linear decrease in egg production with increasing density. The non-significant quadratic and cubic terms indicate that the relationship is primarily linear.
  6. **Model fit:** The overall model explains approximately 72% of the variance in egg production ( $R\text{-squared} = 0.72$ ), indicating a good fit to the data.

## Lecture 12: Writing the Results for a Scientific Paper

Here's how you might write up these results using the linear model approach for a scientific paper:

### Results

A two-way factorial ANOVA revealed that egg mass production in limpets was significantly affected by both adult density ( $F_{3,16} = 9.67$ ,  $P = 0.001$ ) and season ( $F_{1,16} = 17.84$ ,  $P = 0.001$ ), with no significant interaction between these factors ( $F_{3,16} = 0.30$ ,  $P = 0.824$ ). The model explained 72% of the variance in egg production (adjusted  $R^2 = 0.65$ ).

Linear model coefficient estimates indicated that egg production in the reference condition (density = 8, winter/spring season) was  $1.90 \pm 0.17$  (estimate  $\pm$  SE) egg masses per limpet. Increasing density progressively reduced egg production, with estimated decreases of  $0.28 \pm 0.25$ ,  $0.74 \pm 0.25$ , and  $0.91 \pm 0.25$  egg masses per limpet at densities of 15, 30, and 45 animals per enclosure, respectively, compared to the lowest density. Summer/autumn season reduced egg production by  $0.75 \pm 0.18$  egg masses per limpet compared to winter/spring.

Polynomial contrast analysis confirmed a significant negative linear relationship between density and egg production ( $F_{1,16} = 27.58$ ,  $P = 0.001$ ), while quadratic ( $F_{1,16} = 1.29$ ,  $P = 0.272$ ) and cubic ( $F_{1,16} = 0.13$ ,  $P = 0.720$ ) components were not significant. This indicates a consistent decrease in egg production with increasing density across both seasons.

Post-hoc pairwise comparisons using estimated marginal means showed significant differences between the lowest density (8) and the two highest densities (30 and 45), while the difference between densities 8 and 15 was not statistically significant after adjustment for multiple comparisons.

Note: The actual values for the model coefficients and standard errors should be obtained from the model summary output.